

Question	Answer	Marks
4(a)(i)	(normal) force per unit cross-sectional area	B1
4(a)(ii)	extension per unit unstretched length	B1
4(b)(i)	$E = FL / Ax$	C1
	$A = (9.0 \times 0.84) / (1.9 \times 10^9 \times 0.47 \times 10^{-3})$	C1
	$= 8.5 \times 10^{-6} \text{ m}^2$	A1
4(b)(ii)	F , L and x are all the same (for both wires / as in X) or F and strain are the same	B1
	A is greater (for Y), so the Young modulus (for Y) is less than $1.9 \times 10^9 \text{ Pa}$ or less than that of wire X	B1

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